

Exercise 4: Linear systems

Variance in the Ornstein-Uhlenbeck process

The objective of this question is to reproduce figure 5.7 in the book (p. 108). So, consider the scalar linear system

$$\dot{X}_t = -\lambda X_t + \sigma U_t, \quad X_0 = x$$

where $\{U_t : t \geq 0\}$ is Gaussian “white noise”, i.e. the formal derivative of standard Brownian motion. The initial condition $X_0 = x$ is deterministic. For numerical work, we take parameters $x = 1$, $\lambda = 1$, $\sigma = 1$.

Question 1: Write up the mean $\mathbf{E}X_t$ and the variance $\mathbf{V}X_t$ as a function of time (yes, the expressions can be found in the book). Plot the mean, and the mean plus/minus the standard deviation.

Question 2: Assuming that $\lambda > 0$, write up the steady-state variance, $\lim_{t \rightarrow \infty} \mathbf{V}X_t$. Compute its numerical value and compare with the value of $\Sigma(t)$ for t large.

Question 3: Simulate a sample path of $\{X_t\}$ and add it to the graph.

Steady-state variance structure for a mass-spring-damper system

Consider the mass-spring-damper system in the notes (5.1), (5.2), p. 90, with the force $\{F_t : t \geq 0\}$ being white noise with a given intensity $S_{FF}(\omega) = \sigma^2$.

Question 4: Write the system in the standard form $dX_t = AX_t dt + G dB_t$, i.e. specify A and G . *Hint:* The answer is *almost* given in the textbook. Use that standard Brownian motion is the integral of white noise with intensity 1, and remember that intensity - like variance - is a quadratic property.

Question 5: Using the general form, simulate the system on the time interval $t \in [0, 1000]$ using the Euler-Maruyama method. Take system parameters $m = 1$ kg, $k = 0.5$ N/m, $c = 0.2$ Ns/m, $\sigma^2 = 100$ N²s. Let the system start at rest at $t = 0$. Use a time step of $\Delta t = 0.01$ s. Plot the first part of the sample path, $t \in [0, 100]$.

Question 6: Estimate from your simulation the steady-state variance of position Q_t , of velocity V_t , and the covariance between the two. Compare with the solution of the algebraic Lyapunov equation governing the variance. *Note:* In **Matlab**, use built-in `lyap.m`. In **R**, use the function `lyap.R` in **SDEtools**. In **python**, use `scipy.linalg.solve_continuous_lyapunov`.

Question 7: The kinetic energy is $\frac{1}{2}mV_t^2$ while the potential energy is $\frac{1}{2}kQ_t^2$. In steady-state, what is the expected kinetic energy and the expected potential energy? *Note:* The result is an example of equipartitioning of energy, a general principle in statistical mechanics, both quantum and classical.

Question 8: For the simulation, compute and plot the empirical a.c.f. of $\{Q_t\}$ up to lag 50 s. *Hint:* In Matlab and R, use `acf`. In python, use e.g. `statsmodels.tsa.stattools.acf`. Add to the plot the theoretical prediction.

Question 9: Plot, as a function of the frequency ω , the amplitude and phase of the frequency response from the noise to the position. Plot also the theoretical variance spectrum of the position.